

AIN3205: Virtual & Augmented Reality

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Virtual Environments

Lect. 1 Introduction

What is virtual reality (VR)?

- Virtual Reality is a fairly complex technology with few complete off-the-shelf solution bundles.
- Creating a working application often requires a systems approach with fairly broad range of knowledge and talent.

Many Names of VR

- *Virtual Reality*
- *Virtual Environment*
- *Artificial Reality*
- *Computer-Generated Environment*
- *Computer-Simulated Environment*
- *Synthetic Environment*
- *Spatial Immersion*
- *Cyberspace*
- *Virtual Worlds*
- *Virtual Presence*
- *Metaverse*

A simple definition of VR

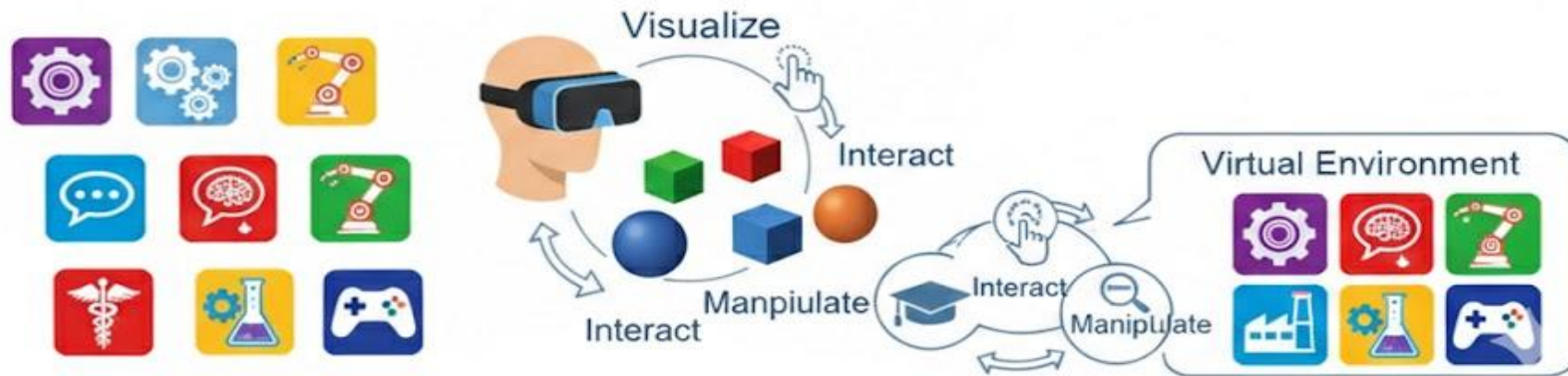
**“A Computer-Generated, 3D Spatial Environment
in Which Users Can Participate in Real-time.”**
Michael Capps, Naval Postgraduate School



Virtual Environments

- ❑ Virtual environment (VE) may be a CAD model, a scientific simulation, or a view into a database.
- The *users can visualize, interact and directly manipulate objects within VE*. Some virtual environments are animated by other processes, physical simulations, or simple animation scripts.
- VE technology has already found *applications* in:
 - industrial design
 - tele-robotics
 - medicine
 - Education
 - Communications
 - scientific research
 - training
 - entertainment

Virtual Environments



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The Three I's of Virtual Reality



Immersion

The sensation of being physically present in a non-physical world. Achieved through sensory occlusion.



Interaction

The ability to manipulate objects and move freely. The world must react to user inputs in real-time.



Imagination

The power to represent impossible physics, CAD data, or molecular structures that don't exist in reality.

Related Technologies

- *Vehicle/Flight Simulators*
- *“Interactive 3D Computer Graphics” (I3D)*
- *CAD*
- *Computer animation/special effects*
- *PC/Video Games*
- *Interactive Images (QuickTime VR)*



Types of VR Systems... [Isdale] (1/2)

1. **Window on World Systems (WoW)**, or Desktop VR.
2. **Video Mapping** : variation of the WoW approach where the user watches a monitor that shows his body's silhouette interaction with the world.
3. **Immersive Systems**: completely immerse the user's personal viewpoint inside the virtual world. These "immersive" VR systems are equipped with a *Head Mounted Display (HMD)*, or a 'Cave' or room in which the viewer stands.

Types of Virtual Reality System

There are different types of VRS:

1. Windows on World(WoW)

- Also called Desktop VR.
- Using a conventional computer monitor to display the 3D virtual world.



Immersive VR



- Completely immerse the user's personal viewpoint inside the virtual 3D world.
- The user has no visual contact with the physical world.
- Often equipped with a Head Mounted Display (HMD).

Immersive VR can be further classified in

- a) Fully – Immersive VR
- b) Semi – Immersive VR
- c) Non- Immersive VR

a) Fully – Immersive VR

- Fully-immersive simulations give users the most realistic experience possible, complete with sight and sound.
- The VR headsets provide high-resolution content with a wide field of view.
- For example: Whether we're flying or fighting , we'll feel like we're really there.



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b) Semi – Immersive VR

- Semi-immersive experiences provide users with a partially virtual environment to interact with.
- This type of VR is mainly used for educational and training purposes and the experience is made possible with graphical computing and large projector systems.

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c) Non- Immersive VR

- Non-immersive VR are often forgotten as an actual type of VR because it's very common in our everyday lives.
- The average video game is technically considered a non-immersive virtual reality experience.

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Types of VR Systems (2/2)

4. **Telepresence** : links remote sensors in the real world with the senses of a human operator. *Applications =>* remote sensors might be located tele-operated robots for fire fighting, space or undersea operations. Surgeons are using instruments on cables having a video camera at the point of operation.
5. **Mixed Reality**: mixes tele-presence and Virtual Reality systems. The computer-generated inputs are merged with tele-presence inputs and/or the users view of the real world. *Applications =>*
 - surgeon's view of a brain surgery is overlaid with images from earlier CT scans and real-time ultrasound.

Telepresence

- A variation of visualizing complete computer generated worlds.
- Links remote sensors in the real world with the senses of a human operator. The remote sensors might be located on a robot. Useful for performing operations in dangerous environments.



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3. Mixed Reality(Augmented Reality)

- The seamless merging of real space and virtual space.
- Integrate the computer-generated virtual objects into the physical world which become in a sense an equal part of our natural environment.



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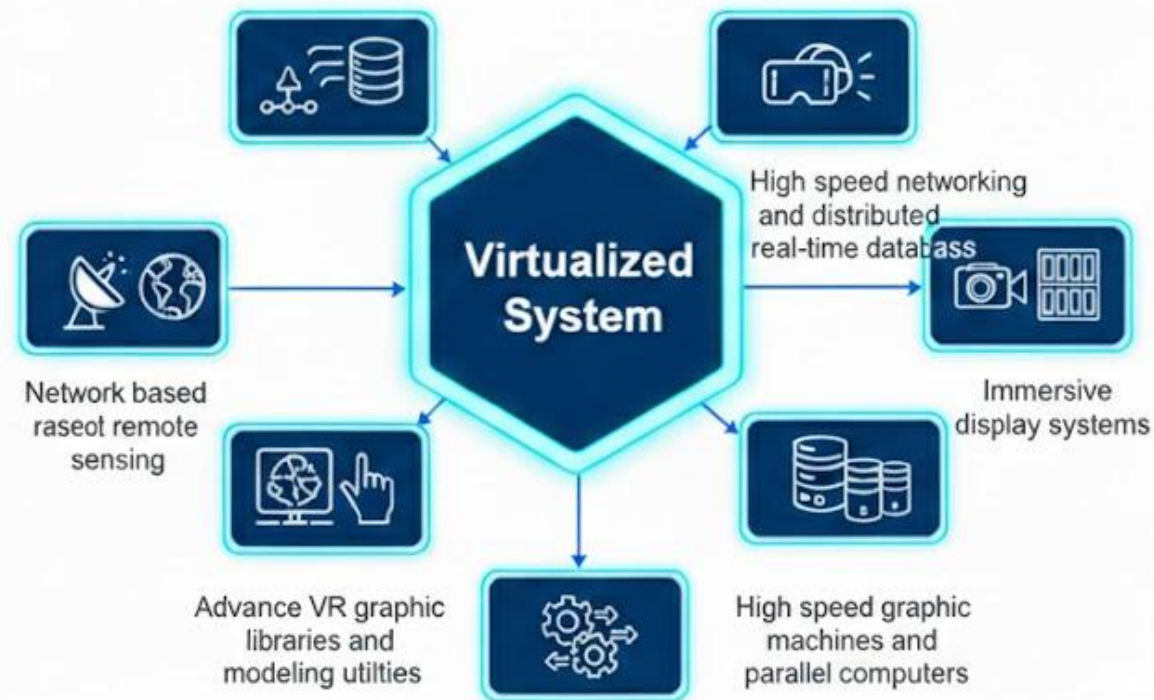
Virtualized Reality Systems

are based on the integration of the following areas
[Boulanger]:-

- High speed networking and distributed real-time database
- Immersive display systems
- Network based remote sensing
- 3D sensors such as: 3D body trackers, color range sensors for short and long distances, and various input devices
- Advance VR graphic libraries and modeling utilities
- High speed graphic machines and parallel computers
- Virtual Simulation Based on Physical Modeling

Virtualized Reality Systems

are based on the integration of the following areas



VR as special case of Interactive 3D Computer Graphics I3D

Both based on:

- Modeling,
- Simulation,
- Interaction,
- Rendering

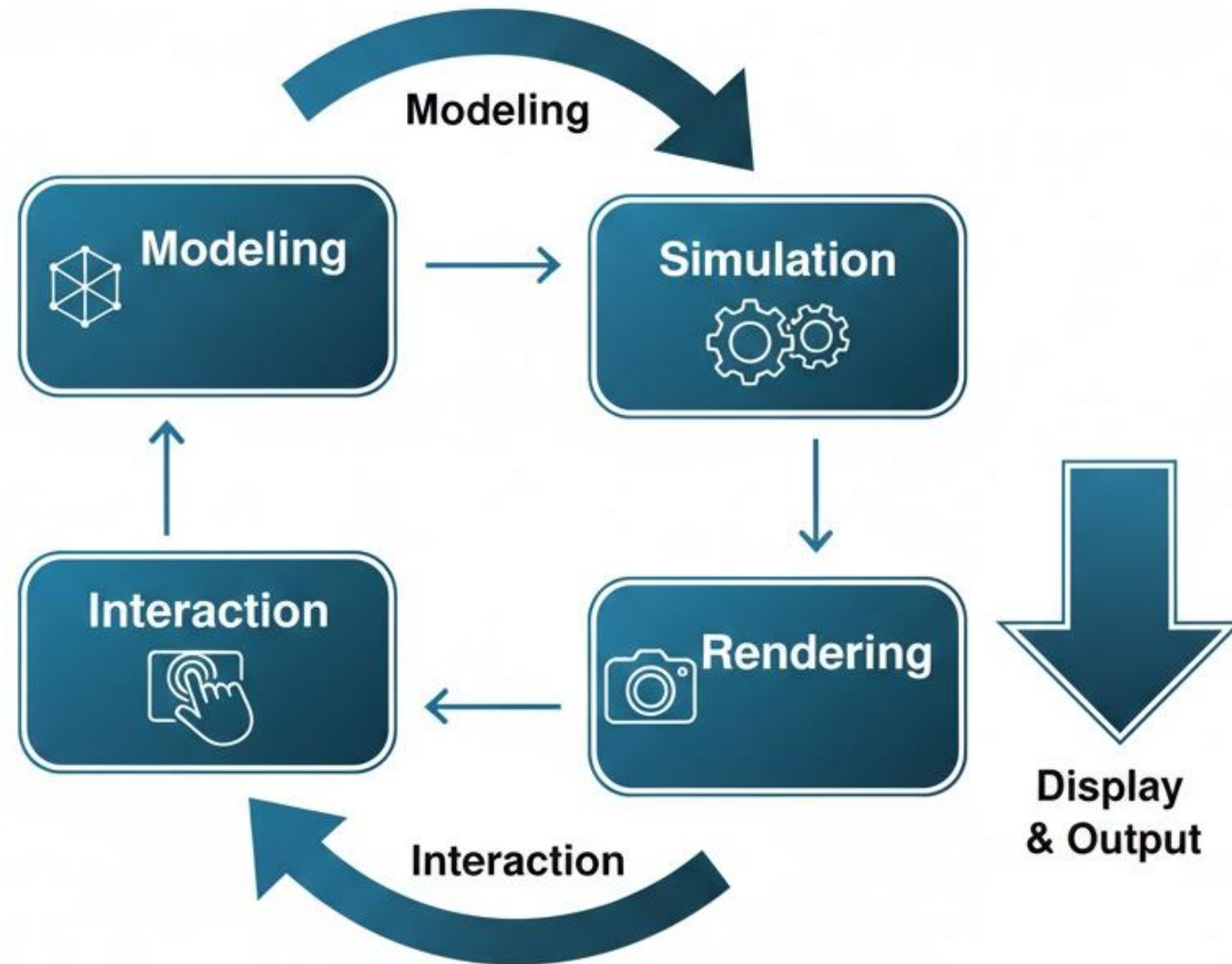


VR is a special case based on:-

- Interface devices & techniques
- Immersion & Presence
- Free range navigation
- Manipulation of world entities
- Other senses (audio, haptics, etc)



Interactive 3D Computer Graphics (I3D)



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Why Not Use VR (barriers)

- *System Costs (equipment, space, people)*
- *Development Complexity*
- *Appropriate Content Design*
- *Cumbersome Equipment*

VE Applications

More details later

- ***Entertainment***
- ***Design and Development***
- ***Training***
- ***Education***
- ***Information / Scientific Visualization***
- ***Collaboration / Community***
- ***Medicine / Mental Health***
- ***Marketing***



Augmented Reality & Computer Augmented Environments (1/3)

<http://www.cs.rit.edu/~jrv/research/ar/>

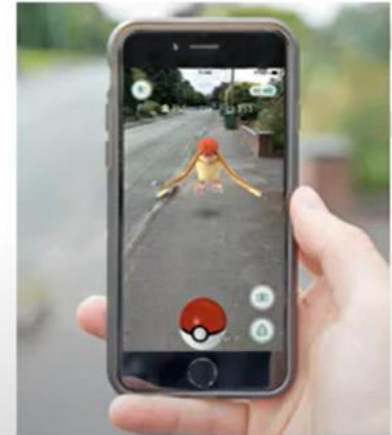
- “The basic difference between the Augmented Reality and the Virtual Reality is the **immersiveness** of the system.
- Virtual reality strives for a totally immersive environment. The visual, and in some systems aural and receptive senses are under control of the system.
 - Augmented reality system augments the real-world scene necessitating that the user maintains a sense of presence in that world. The virtual images are merged with the real view to create the augmented display. There must be a mechanism to combine the real and virtual” *[Valino]*.

Augmented reality (AR)

- Augmented reality (AR) is a technology that layers computer-generated enhancements atop an existing reality.
- AR is developed into apps and used on mobile devices to blend digital components into the real world.
- AR technology is used to display score overlays on telecasted sports games and pop out 3D emails, photos or text messages.
- Leaders of the tech industry are also using AR to do amazing and revolutionary things with holograms and motion activated commands.

Augmented reality (AR)

- ❖ A sister technology to VR is **augmented reality (AR)**, which combines computer-generated imagery (CGI) with views of the real world.
- ❖ AR on smartphones has recently garnered widespread interest with **Vuforia** AR toolkit which is integrated directly with the Unity game engine.
- ❖ AR on a mobile device overlays the CGI on top of live video from a camera.





- Augmented reality and virtual reality are inverse reflections of one in another with what each technology seeks to accomplish and deliver for the user.

Augmented Reality & Computer Augmented Environments (3/3)

“Tele-Immersion will enable users at geographically distributed sites to collaborate in real-time in a shared, simulated environment as if they were in the same physical room.

This paradigm for human-computer interaction is the ultimate synthesis of networking and media technologies and, as such, it is the greatest technical challenge for Internet”.

| Interacting with Digital Worlds

Sensory Perception

How the environment "speaks" to the human user:

- **Visual:** Stereoscopic depth and wide FOV.
- **Aural:** Spatialized 3D audio cues.
- **Haptic:** Force feedback and tactile vibration.

Motor Control

How the human user "speaks" to the environment:

- **Head Tracking:** Look around naturally.
- **Gestural:** Use hands to grab or point.
- **Locomotion:** Walk, teleport, or navigate.

Defining VR and Virtual Environments

Virtual Reality (VR)

The total system—hardware and software—facilitating human immersion. It encompasses tracking sensors, computational units, and sensory displays.

Example: A flight simulator cockpit combined with the motion platform and visual rendering engine.

Virtual Environment (VE)

The computer-generated model itself. It is the synthetic world containing geometry, physics, and interactive logic.

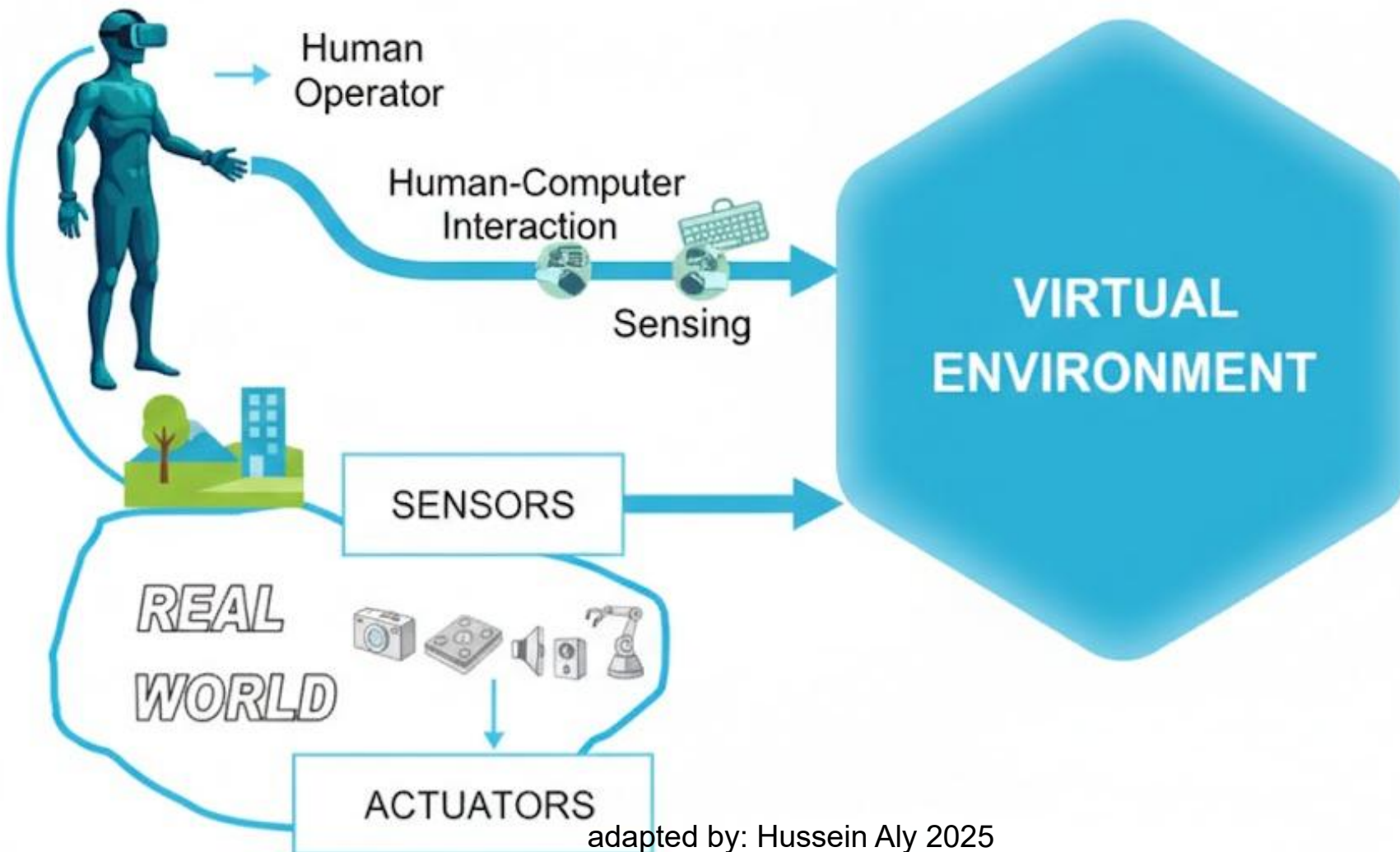
Example: The 3D mathematical model of the terrain, weather, and airport used within the flight simulator.

Interfacing VE with Human Operators

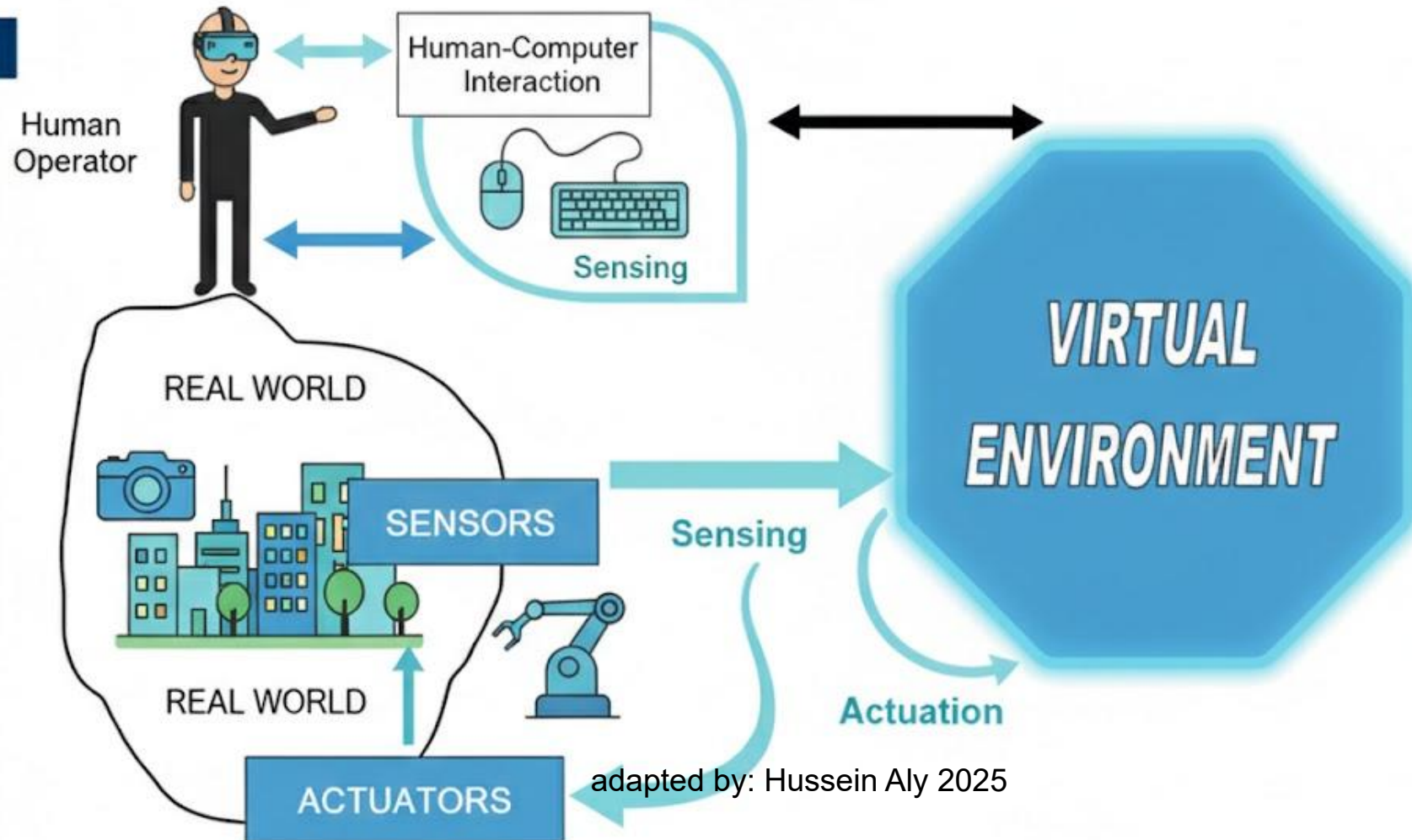
Interface Component	Functionality	Technology Examples
Sensing (Input)	Captures human motion/state	IMUs, Optical Tracking, Eye Trackers
Actuation (Output)	Stimulates human senses	OLED Displays, Spatial Audio, Haptics
Control Loop	Processes user action in VE	Physics Engines, Collision Detection
Feedback	Force/Tactile response	Vibrotactile vests, Haptic Gloves

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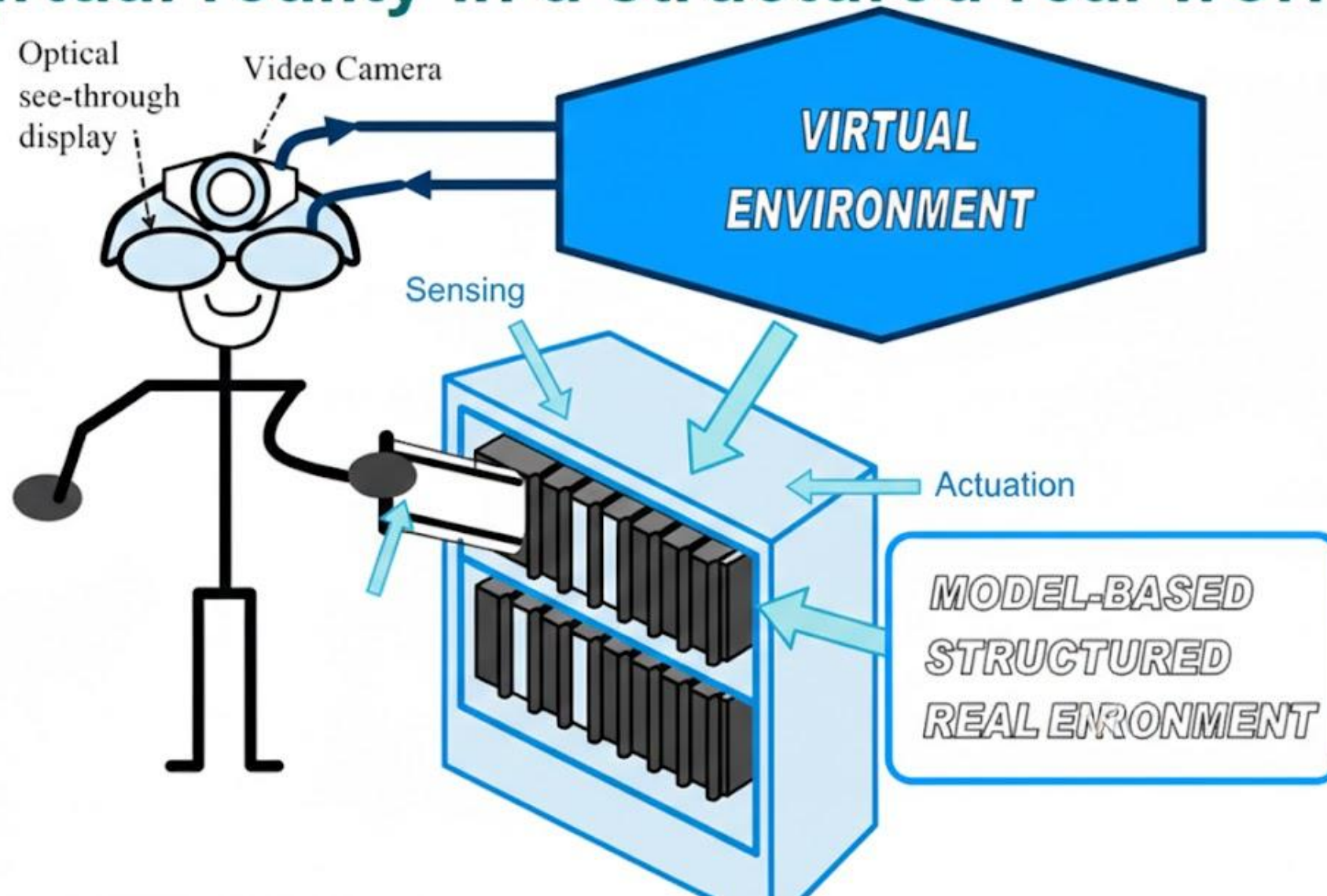
Interfacing virtual environments with the real world and human operators



Human interaction with the real world and VE in augmented virtual reality applications



Human operator using augmented virtual reality in a structured real world



The Operator in a Structured World



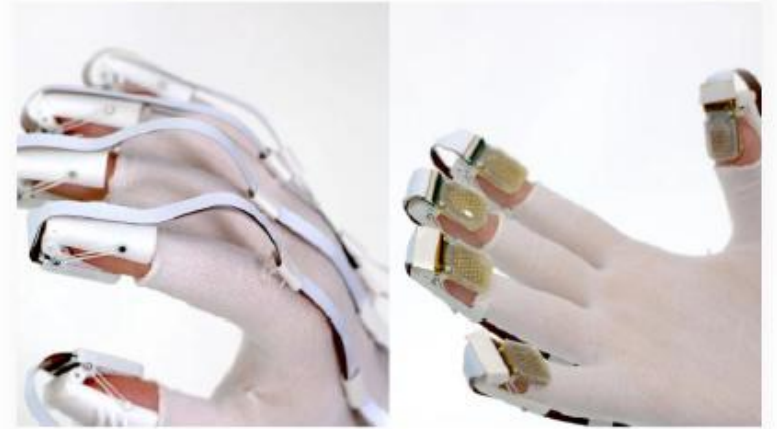
Medical AVR

Surgeons using 3D MRI overlays during live operations to guide incisions.



Collaborative Design

Engineers interacting in a CAVE to evaluate vehicle aerodynamics.



Remote Handling

Operators using haptic interfaces to control robots in hazardous areas.

VE Technology

- Programmatics
- VR System Components



VR System Components

Hardware:

- *Input Devices,*
- *Displays*
- *Processors*

Software:

- *Simulation Manager*
- *World Database*
- *Authoring*
- *Interaction Techniques*
- *Application Software*
- **Network Connectivity (Optional)**



THANK YOU

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